WEEK 1 · DAY 4 · OVERVIEW

# Classes

## *A Bond Object*




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ABOUT 50 MIN WITH EXERCISES / 8 SECTIONS / 6 EXERCISES

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# The Desk Lead's Sixth Note

Every day we hand `current_yield` a coupon and a price, and on Day 3 a price of zero walked straight in. Build me a Bond that carries its own terms, works out its own numbers, refuses a bad price, and prints in words. Then tie the book's value and income from four Bonds.

ABOUT 50 MIN WITH EXERCISES / 8 SECTIONS / 6 EXERCISES

# Your Route Through Day 4

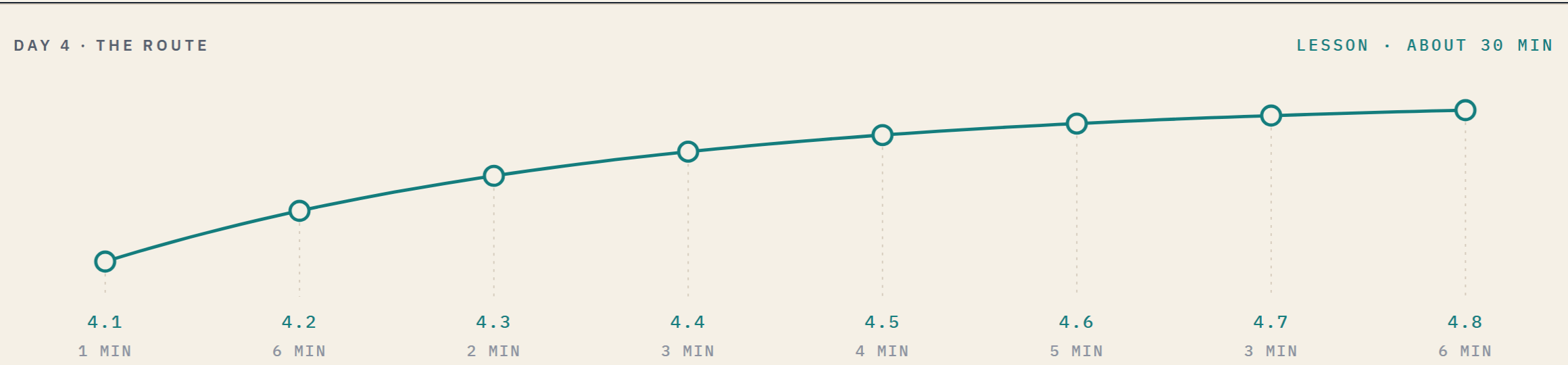



FIGURE 2 Eight sections, in order, with about how long each takes.

**Time. About 30 minutes for the lesson, then about 18 for the six exercises.**

# 4.1 Objects Are Already Everywhere

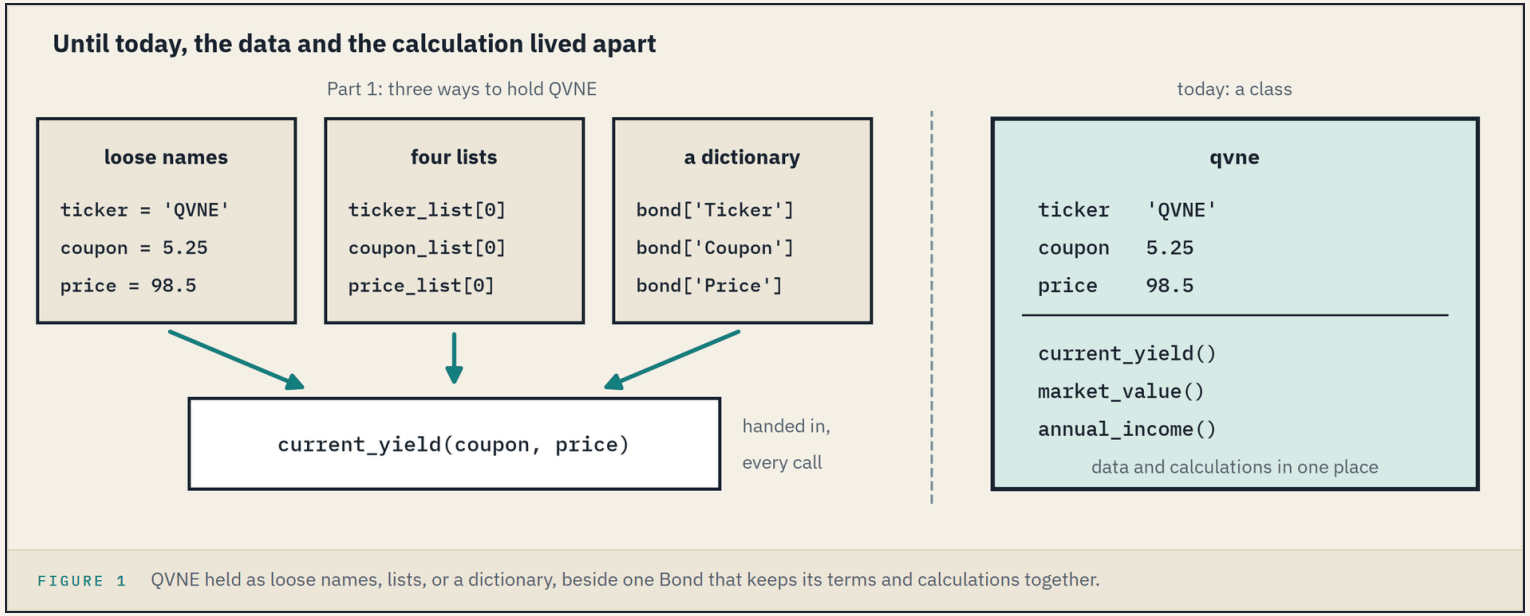



FIGURE 1 QVNE held as loose names, lists, or a dictionary, beside one Bond that keeps its terms and calculations together.

THE CODE

```
coupon = 5.25
ticker = 'QVNE'
print(type(coupon))
print(type(ticker))
print(ticker.lower())
```

```
<class 'float'>
<class 'str'>
qvne
```

**Python's own classes cover numbers and text. Now write one for a bond.**

# 4.2 A First Class

A FIRST CLASS

```
class Bond:
    """A bond: its terms, and its current yield."""
    def set_terms(self, ticker, coupon, price):
        self.ticker = ticker
        self.coupon = coupon
        self.price = price
    def current_yield(self):
        return self.coupon / self.price * 100
```

MAKE ONE, FILL IT IN

```
qvne = Bond()
qvne.set_terms('QVNE', 5.25, 98.5)
print(type(qvne))
```

```
<class '__main__.Bond'>
```

ASK IT

```
print(qvne.coupon)
print(qvne.current_yield())
```

```
5.25
5.32994923857868
```

**A class is a blank form until we fill one in.**

# What self Means

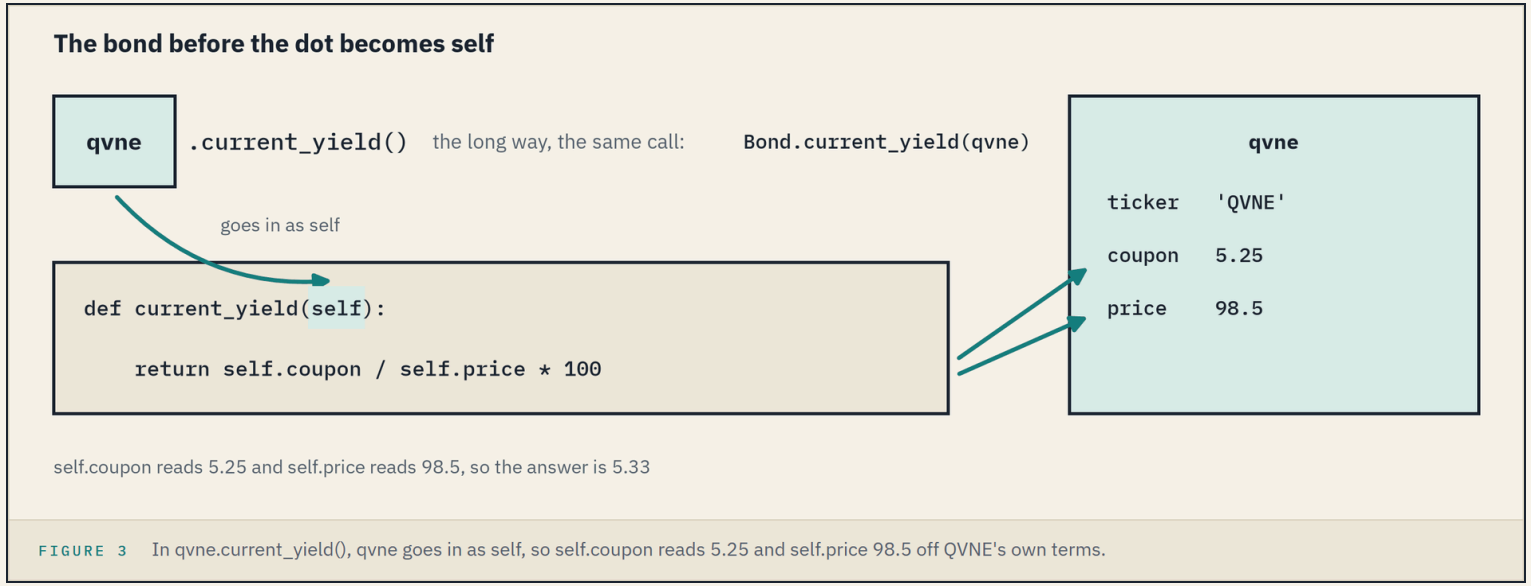



FIGURE 3 In qvne.current_yield(), qvne goes in as self, so self.coupon reads 5.25 and self.price 98.5 off QVNE's own terms.

THE LONG WAY

```
print(Bond.current_yield(qvne))
5.32994923857868
```

OUTSIDE THE CLASS

```
print(self.coupon)
NameError: name 'self' is not defined
```

**Inside the class `self` is this bond; outside, use the bond's own name.**

# 4.3 The Trouble With Loose Attributes

A SECTOR ON ONE BOND ONLY

```
qvne.sector = 'Energy'
print(qvne.sector)
```

```
Energy
```

THE NEXT BOND

```
print(dnmr.sector)
```

```
AttributeError: 'Bond' object has no attribute 'sector'
```

A BOND NEVER FILLED IN

```
plwk = Bond()
print(plwk.coupon)
```

```
AttributeError: 'Bond' object has no attribute 'coupon'
```

**Every Bond needs the same terms from the moment it is made.**

# 4.4 The Initializer

**One class, four objects: the terms differ, the methods are shared**

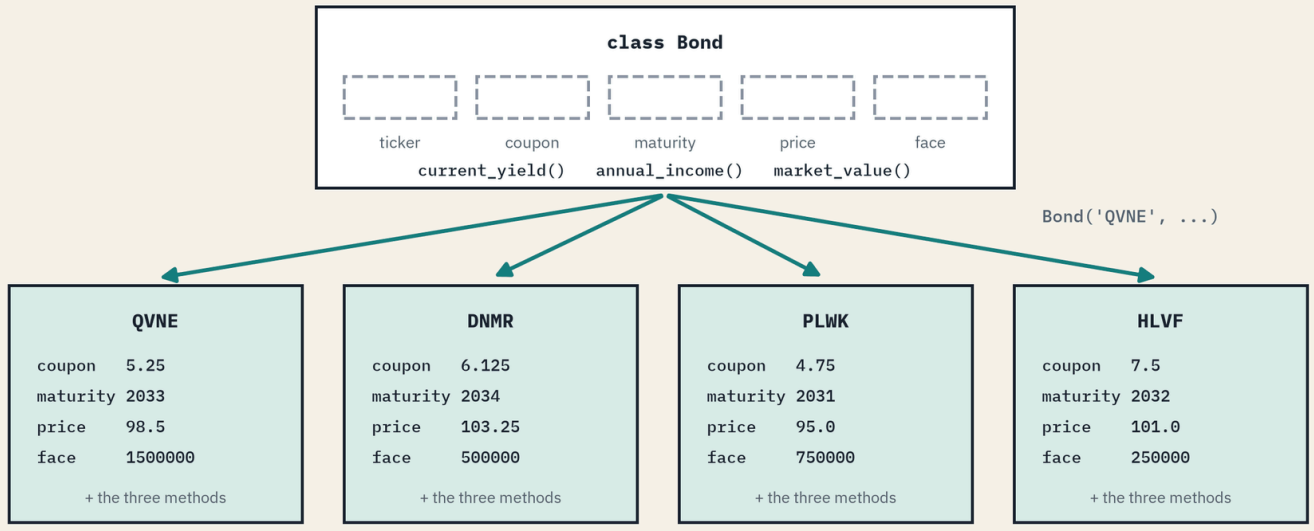



FIGURE 4 One Bond class, five terms and three methods, makes four objects, each with its own terms and the same methods.

FOUR BONDS, ONE LINE EACH

```
qvne = Bond('QVNE', 5.25, 2033, 98.5, 1500000)
dnmr = Bond('DNMR', 6.125, 2034, 103.25, 500000)
plwk = Bond('PLWK', 4.75, 2031, 95.0, 750000)
hlvf = Bond('HLVF', 7.5, 2032, 101.0, 250000)
print(plwk.coupon)
```

```
4.75
```

**The initializer will not make a half-filled Bond.**

# 4.5 Working With an Object

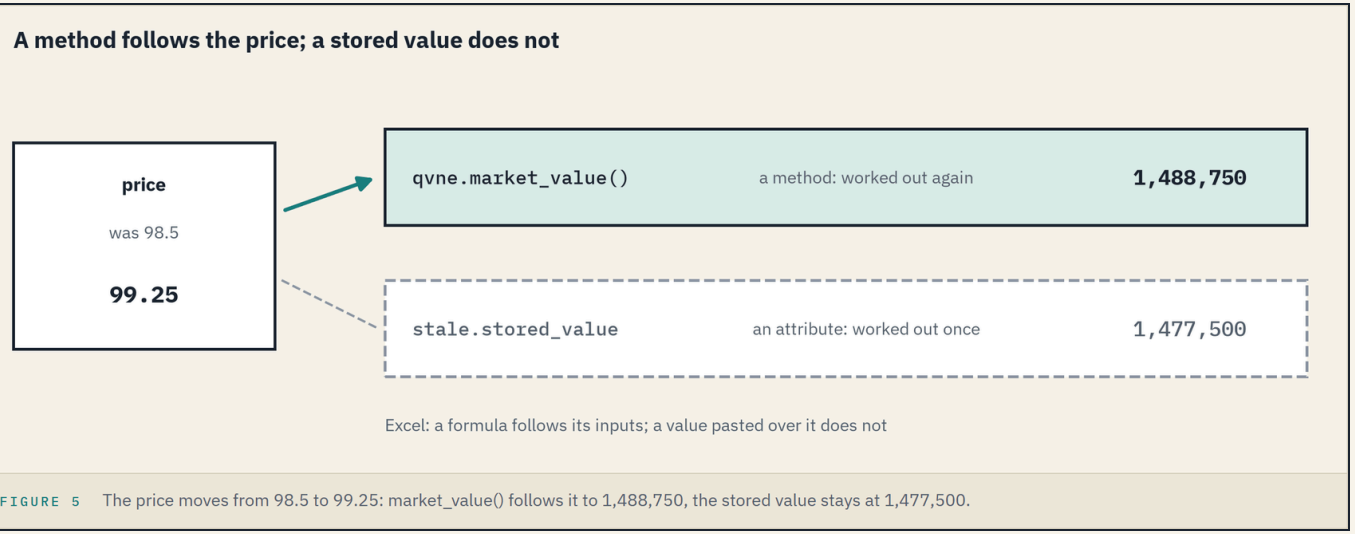



FIGURE 5 The price moves from 98.5 to 99.25: market_value() follows it to 1,488,750, the stored value stays at 1,477,500.

### THE METHOD FOLLOWS THE PRICE

```
print(f'Market value: ${qvne.market_value():,.2f}')
```

```
Market value: $1,477,500.00
```

### A STORED VALUE GOES STALE

```
stale = BondWithStoredValue('QVNE', 98.5, 1500000)
print(f'Before the mark: ${stale.stored_value:,.2f}')
print(f'After the mark:  ${stale.stored_value:,.2f}')
```

```
Before the mark: $1,477,500.00
After the mark:  $1,477,500.00
```

**Store the facts as attributes; calculate anything that depends on them in a method.**

# 4.6 A Bond That Refuses a Bad Price

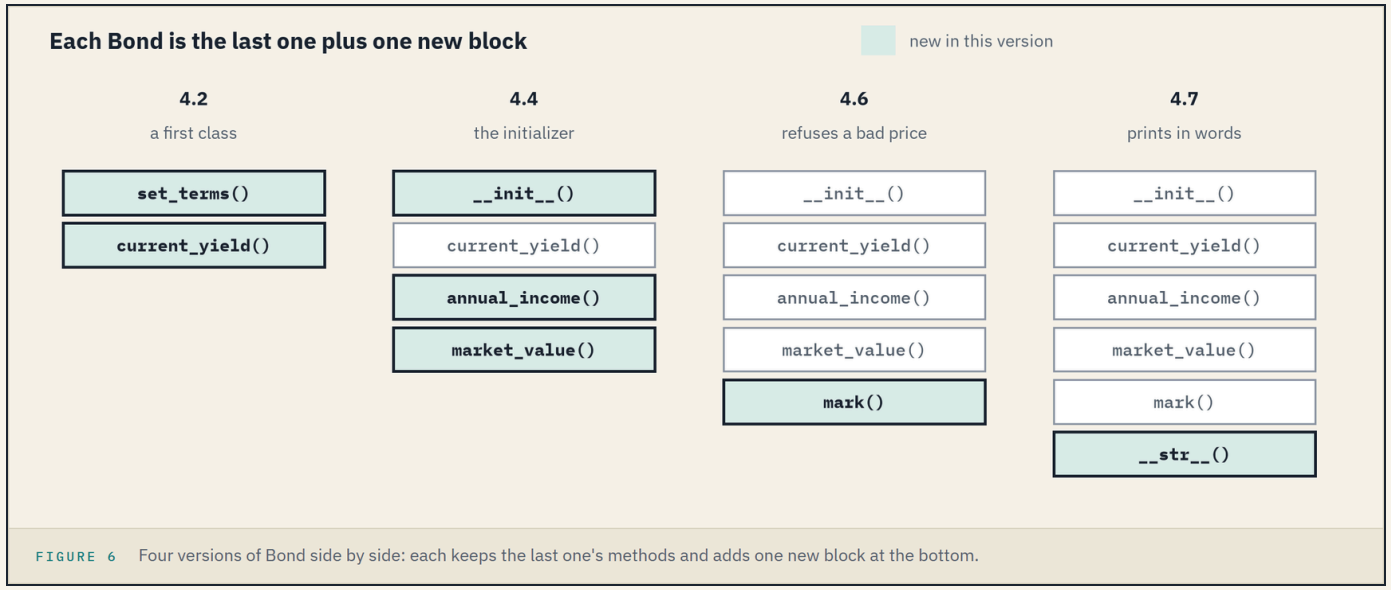



FIGURE 6 Four versions of Bond side by side: each keeps the last one's methods and adds one new block at the bottom.

### MARK IT, THEN TRY ZERO

```
qvne.mark(99.25)
qvne.mark(0)
```

```
ValueError: the price must be above zero, and it is 0
```

### THE REFUSED PRICE NEVER ARRIVED

```
print(qvne.price)
```

```
99.25
```

**The Bond refuses a bad price itself: Day 3's `raise`, now inside the class.**

# 4.7 Printing an Object

### WHAT PRINT SHOWS

```
print(qvne)
```

```
<__main__.Bond object at 0x0000023387F2ABA0>
```

### __STR__ GIVES IT WORDS

```
qvne = Bond('QVNE', 5.25, 2033, 98.5, 1500000)
print(qvne)
```

```
QVNE 5.25% of 2033, current yield 5.33%
```

### PYTHON CALLS THESE ITSELF

| METHOD | PYTHON CALLS IT WHEN |
|---|---|
| `__init__` | a Bond is made |
| `__str__` | a Bond is printed |
| `__repr__` | a list of Bonds is printed; not written today |

### EXCEL SIDE BY SIDE

**Excel side by side.** `__str__` builds a label from the facts, as a formula does: with ticker, coupon, and maturity in A2:C2, `=A2&" "&B2&"% of "&C2` returns QVNE 5.25% of 2033.

**Now a Bond prints in words.**

# 4.8 A Portfolio of Objects

**A list of Bonds reads like a holdings sheet**

attributes: plain columns — methods: formula columns

the class →

| Ticker | Coupon | Maturity | Price | Face | Current yield | Market value | Income |
|---|---|---|---|---|---|---|---|
| QVNE | 5.25 | 2033 | 98.5 | 1,500,000 | 5.33 | 1,477,500 | 78,750 |
| DNMR | 6.125 | 2034 | 103.25 | 500,000 | 5.93 | 516,250 | 30,625 |
| PLWK | 4.75 | 2031 | 95.0 | 750,000 | 5.00 | 712,500 | 35,625 |
| HLVF | 7.5 | 2032 | 101.0 | 250,000 | 7.43 | 252,500 | 18,750 |

each row: one object

fill down = the loop

FIGURE 7 The four bonds as a sheet: plain columns are attributes, formula columns are methods, each row is an object.

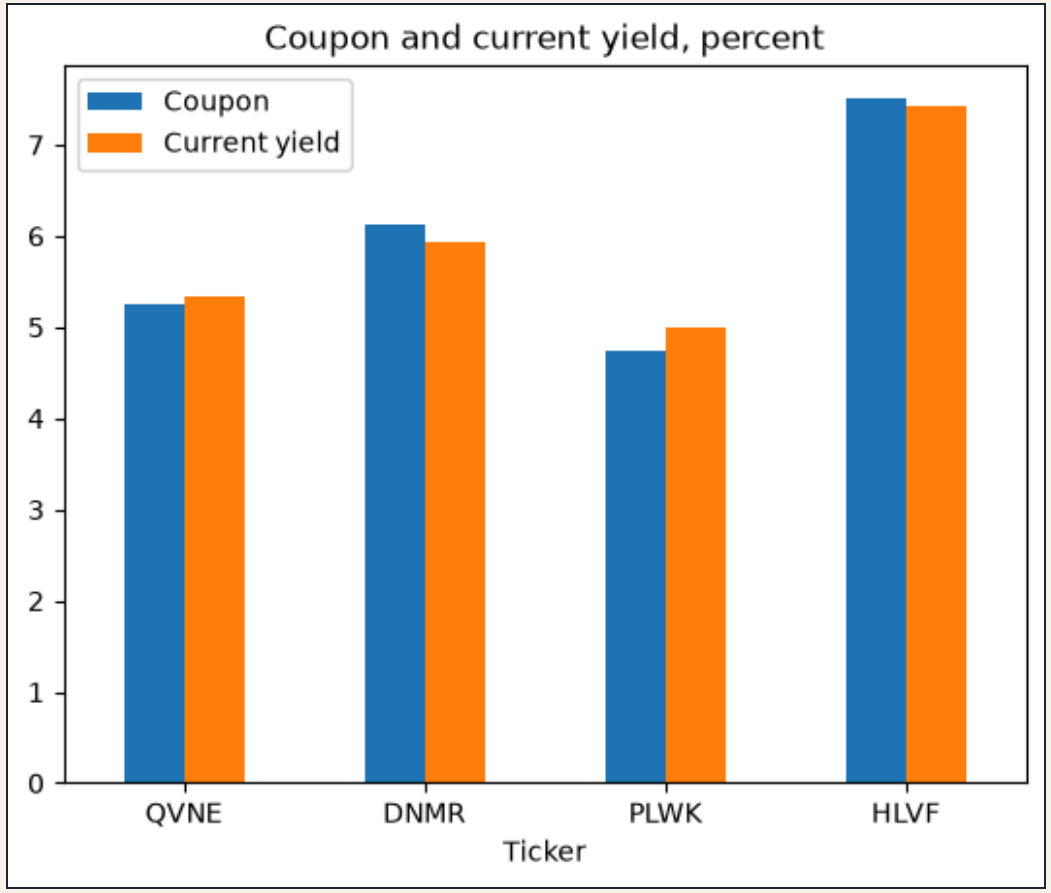
THE BOOK IS A LIST

```
portfolio = [qvne, dnmr, plwk, hlvf]
for bond in portfolio:
    print(bond)
```

```
QVNE 5.25% of 2033, current yield 5.33%
DNMR 6.125% of 2034, current yield 5.93%
PLWK 4.75% of 2031, current yield 5.00%
HLVF 7.5% of 2032, current yield 7.43%
```

**Each bond carries everything it needs.**

# Discounts and the Book




EACH BOND, ASKED

| | TICKER | COUPON | CURRENT YIELD |
|---|---|---|---|
| 0 | QVNE | 5.250 | 5.329949 |
| 1 | DNMR | 6.125 | 5.932203 |
| 2 | PLWK | 4.750 | 5.000000 |
| 3 | HLVF | 7.500 | 7.425743 |

THE BOOK TIES

```
book_value = sum([bond.market_value() for bond in portfolio])
book_income = sum([bond.annual_income() for bond in portfolio])
```

```
Book ties: $2,958,750.00 value, $163,750.00 income
```

**Below par, the current yield is above the coupon; above par, below it.**

# You Can Now

- Read a class: its attributes, its methods, and **`self`**.
- Make a **Bond** in one line with an initializer.
- Store the facts as attributes and calculate the rest in methods.
- Make a method refuse a bad value with **`raise`**.
- Make an object print in words a person can read.
- Loop over a list of objects.

| THE NOTE ASKED | ANSWER |
|---|---|
| Its own terms and numbers | `current_yield()`, `market_value()`, `annual_income()`, nothing handed in |
| Refuse a bad price | `qvne.mark(0)` stops with a **`ValueError`** that says why |
| Print in words | `QVNE 5.25% of 2033, current yield 5.33%` |
| The book from four Bonds | two **`assert`** lines tie the value and the income |

NEXT
WEEK 1 · DAY 5

## Case Study: Organizing a Small Portfolio

One job from start to finish, using everything from the week.